

Water consumption by ALL data centres for ALL uses (AI included, training included, and all other data centres included), globally

Posted by u/Ohigetjokes • 0 points • 40 comments

To address the cope and whataboutism I dug up a slightly more detailed chart than the hamburger one.

Comments

u/Daflehrer1 • 1 points • 2025-10-16 20:32 UTC

What is the source of this graph, and how old is it? I ask because data center are being planned and built literally every day.

u/Ch33seBurg • 1 points • 2025-08-27 18:15 UTC

I'm not trying to sound bias or anything, but that stuff is very necessary for all of us...

u/Ohigetjokes • 1 points • 2025-08-27 19:07 UTC

True but... you see the minuscule amount of water usage, right? And now a 1% efficiency increase in literally any sector would more than account for data center usage, right?

u/crossorbital • 2 points • 2025-08-24 22:24 UTC

Yeah, but how are you supposed to rake in imaginary internet points and boost engagement metrics by complaining about farmers? You can't grift off things that aren't trendy, it's just not practical.

u/Worldly_Air_6078 • 1 points • 2025-08-24 20:02 UTC

Exactly. As a vegan, I'll take criticism about the water used by AI only from other vegans. One hamburger corresponds to 2400 liters of water, it would take hundreds of hours of incessant use of AI to consume a noticeable fraction of this quantity of water.

u/realechelon • 2 points • 2025-08-30 16:33 UTC

Even then, you'd want the subset of vegans who avoid high water usage plants like avocados, certain nuts & legumes etc. Not all plant foods are water-friendly (even though none come close to cattle).

I'm not a vegan but I avoid beef & dairy for environmental reasons.

u/Educational-Sun-7902 • -1 points • 2025-08-24 19:48 UTC

source?

u/GoldAttorney5350 • 20 points • 2025-08-24 17:51 UTC

There has to be some sort of astroturfing going on in the anti AI movement. I've never seen a topic where so many people truly believe in so much misconceptions or downright disinformation. It's like there's a deliberate attempt to frame AI as this hugely horrible thing to ever come out, and any attempt to prove otherwise literally makes you a target of cancellation and harassment.

u/UnknownTreeBears • 1 points • 2025-08-25 19:44 UTC

No, there are just a lot of us who have seen this whole song and dance from the rich and powerful before and we know what a bunch of bullshit it actually is. This is the dotcom bubble all over again and if you aren't old enough to remember that then it explains a whole lot.

We're not buying into hype and recognizing the very real harm that comes from AI. I also happen to know enough about computers and brains to know how much smoke they're blowing up our asses with all these promises; and I fundamentally don't trust anything that a bunch of megacorps are trying so very hard to sell me.

u/insidiouspoundcake • 3 points • 2025-08-25 01:45 UTC

I genuinely think that China would be stupid (on a statecraft level) not to be bare minimum signal-boosting misinfo, if not outright botting and astroturfing on this topic. Machine learning and neural networks are obviously going to be a big deal at scale, regardless of the specifics of current LLM architecture.

u/crossorbital • 9 points • 2025-08-24 22:29 UTC

If there is astroturfing, I'd guess it's the big media corporations. The shitstorm only really got started when local image generation became useful, and like 90%+ of what antis seem to want would conveniently benefit the big IP holders one way or another.

That said, don't underestimate people's ability to be deeply uninformed and very angry about it all on their own.

u/Scam_Altman • 5 points • 2025-08-24 21:19 UTC

>I've never seen a topic where so many people truly believe in so much misconceptions or downright disinformation.

Flat earthers and antivaxxers.

u/GoldAttorney5350 • 4 points • 2025-08-25 02:18 UTC

But this is way more than that. I mean you'll probably never encounter a sincere flat earther in real life, but I think you'd know at least 1 or 2 anti AI people

u/Scam_Altman • 3 points • 2025-08-25 02:43 UTC

>I mean you'll probably never encounter a sincere flat earther in real life

In 2020, it was reported that based on polling by Datafolha, 7% of Brazilians believed in a flat Earth.[121] A 2018 YouGov poll found that around 4% of the population of the United States believed in flat Earth,[122] while the POLES 2021 Survey found around 10% of the United States population believed that Earth is flat.[123] A 2019 YouGov survey found that around 3% of British people supported flat Earth.[124]

https://en.m.wikipedia.org/wiki/Modern_flat_Earth_beliefs#:~:text=A%202018%20YouGov%20poll%20found%20that%20around,3

u/Thatunkownuser2465 • 6 points • 2025-08-24 17:22 UTC

Now if we convert how many Lake Meads to use:

Its **maximum capacity** is about **35 km³** of water.

* **Agriculture (2,900 km³)** → $2,900 \div 35 \approx 83$ Lake Meads per year

* **Industry (600 km³)** → $600 \div 35 \approx 17$ Lake Meads

* **Municipal (500 km³)** → $500 \div 35 \approx 14$ Lake Meads

* **Cattle (192 km³)** → $192 \div 35 \approx 5.5$ Lake Meads

* **All Data Centers (0.9 km³)** → $0.9 \div 35 \approx 0.026$ Lake Meads — that's about **1/40th** of a single Lake Mead, or basically a **small fraction of one big reservoir**.

u/Ohigetjokes • 2 points • 2025-08-24 19:11 UTC

Idk I think taking a number that itself was an approximation boiled down from much larger numbers and doing all these conversions... gross consumption data from data centers is available

u/NeitherDrummer777 • 6 points • 2025-08-24 17:03 UTC

You can't just look at the amount of water consumed. You also need to differentiate between blue, green and gray water and look at the condition the water is in after usage

Water used in data centers is blue water, which is the most important kind of water but it also leaves in a good condition. It's basically just heated up afterwards and becomes usable again very easily

This is very different for water used on meat, while it's green water (which is better) it leaves the system for quite some time and requires much more effort to become usable again

u/DaveSureLong • 1 points • 2025-08-25 17:36 UTC

You're pretty spot on. I'll add onto this and say the per user usage of water and power is also extremely miniscule compared to per user food items(any).

u/NatureKas • 28 points • 2025-08-24 16:00 UTC

Before someone defends that agriculture is necessary, at least in the USA 30-40% of food gets wasted. We could literally cut agriculture by like 10% and we would still have too much food. There are better ways to save water

u/DaveSureLong • 1 points • 2025-08-25 17:33 UTC

Agriculture is necessary dude. You gotta eat and the more food there is the cheaper it is. Only a moron thinks that food is free.

There is NEVER a situation in real life where you have too much food in truth its always too little

u/Scam_Altman • 15 points • 2025-08-24 19:16 UTC

>Before someone defends that agriculture is necessary

I'ma stop you right there. Animal agriculture is one of the most water intensive and environmentally damaging activities on earth, and completely unnecessary. Cutting beef out of your diet is orders of magnitude more impactful than avoiding AI use.

u/Researcher_Fearless • 2 points • 2025-08-25 00:09 UTC

I mean, if we all stopped eating beef, that would probably cut plant agriculture by something like ten percent all on its own (I am not educated enough on the topic to give a number)

u/Scam_Altman • 0 points • 2025-08-25 01:25 UTC

Based on the analysis provided, here is a response to the conversation:

The estimate that stopping beef consumption would cut plant agriculture by "something like ten percent" is an underestimate. The data indicates the impact would be substantially larger.

Economic modeling suggests that eliminating beef production in developed countries could reduce total crop production needs by 20-40%. Research from Chatham House further specifies that if high-income countries replaced all animal products with plant-based foods, it could result in a 29% reduction in the associated global cropland use.

This significant impact is rooted in the inefficiency of beef production and how agricultural resources are currently allocated:

* Crop Allocation for Feed: A large portion of major crops is grown for livestock, not direct human consumption. In the U.S., approximately 40% of corn and 65% of soybeans are used for animal feed.

* Inefficient Caloric Conversion: Beef production is shown to have a caloric conversion efficiency of only about 3%. It takes roughly 25 calories of feed crops to produce just 1 calorie of beef.

* High Feed Requirements: The feed conversion ratio for beef is greater than 10 kg of feed for every 1 kg of consumable meat produced.

In summary, a large segment of plant agriculture is driven by the demand for animal feed, and beef is a particularly inefficient method of converting those plant resources into human food. Therefore, a reduction in beef consumption would lead to a correspondingly large decrease in the demand for agricultural crops.